Design Science Research

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DSR in Information Systems

Guidelines for Conducting Design Science Research in Information Systems

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Abstract. Information Systems (IS) as a discipline is still young and is continuously involved in building its own research knowledge base. Design Science Research (DSR) in IS is a research strategy for design that has emerged in the last 16 years. IS researchers are often lost when they start with a project in DSR, especially young researchers. We identified a need for a set of guidelines with supporting reference literature that can assist such novice users of DSR. We identified major themes relevant to DSR and proposed a set of six guidelines for the novice researcher supported with references summaries of seminal works from the IS DSR literature. We believe that someone new to the field can use these guidelines to prepare him/herself to embark on a DSR study.

Guidelines for conducting DSR

Table 3. Guidelines for the novice researcher/postgraduate student

Guidelines
1. Contextualise DSR in the field of Information Systems and be able to distinguish between concepts such as design, design science and DSR.
2. Understand the philosophical underpinning of research and the discourse on the nature of DSR.
3. Obtain a historical perspective of DSR and consult the work of the pioneers in the field.
4. Consider the role of the artefact in DSR and the different views on design theory.
5. Select an appropriate DSR method for execution of the research study.
6. Strategise on how research done in DSR should be communicated in a report such as a thesis.

Guideline 1: *Contextualise DSR in the field of Information Systems and be able to distinguish between concepts such as design, design science and DSR.*

- Gregory [14] argues that in doing design one is creating something that does not exist.
- There are two concepts of importance in this argument – there is creation of something (the artefact) and there is the process of creation.
- Design is therefore both a 'noun and a verb' [14:3] or a process and a product.

Guideline 1: ***Contextualise DSR in the field of Information Systems and be able to distinguish between concepts such as design, design science and DSR.***

- In DSR we differentiate between *design* and a *design theory:*
  - design focuses on the 'use of scientific principles, technical information and imagination in the definition of a structure, machine or system to perform pre-specified functions with the maximum economy and efficiency'
  - design theory is 'a prescriptive theory based on theoretical underpinnings which says how a design process can be carried out in a way which is both effective and feasible' [2:36-37].

Guideline 1: ***Contextualise DSR in the field of Information Systems and be able to distinguish between concepts such as design, design science and DSR.***

- Cross [3:66] referred to DS as an 'explicitly organized, rational and wholly systematic approach to design'
- Bayazit [4] focused on the concept of man-made things when he defined design research as a 'systematic inquiry whose goal is knowledge of, or in, the embodiment of configuration, composition, structure, purpose, value, and meaning in man-made things and systems'
- Hevner [5:76] focuses more on the practical nature of DSR when he refers to design science as 'fundamentally a problem solving paradigm'.

Guideline 1: ***Contextualise DSR in the field of Information Systems and be able to distinguish between concepts such as design, design science and DSR.***

- DSR in IS reached a milestone in 2004 with Hevner et al. [5] presenting their framework for IS research and guidelines for DSR. In this work they referred to DSR as a paradigm where the "knowledge and understanding of a problem domain and its solution are achieved in the building and application of the designed artefact" [5:75].
- More or less in the same timeframe Vaishnavi, Keuchler and Petter [17] started a web site focusing on DSR in IS. According to them, DSR uses a set of synthetic and analytical techniques and perspectives for performing research in IS.

Guideline 2: *Understand the philosophical underpinning of research and the discourse on the nature of DSR.*

- Research conducted in IS is mostly multi-disciplinary and the philosophy mostly used is either positivist, interpretivist or critical research.
- In the papers reviewed, three discourses emerged, including:
  1. DSR as paradigm
  2. Traditional paradigms and
  3. Pragmatism paradigm.

Guideline 2: *Understand the philosophical underpinning of research and the discourse on the nature of DSR.*

1. DSR as paradigm

- Originally Vaishnavi, Keuchler and Petter [17] discussed DSR as a paradigm on its own. They argued that design can be research and that it changes the world through the development of new artefacts.
- Their initial ideas were shared on a website hosted by DESRIST and later replicated in a book [18] in which they contrast interpretivism, positivism and DSR in tabular format.

Table 4. Philosophical assumption of DSR [18]

Ontology	Epistemology	Methodology	Axiology
Multiple, contextually situated alternative world-states. Socio-technologically enabled	Knowing through making: objectively constrained construction within a context. Iterative circumscription reveals meaning	Developmental. Measure artefactual impacts on the composite system	Control; creation; problem-solving; progress (i.e., improvement); understanding

Guideline 2: *Understand the philosophical underpinning of research and the discourse on the nature of DSR.*

2. Traditional paradigms

- In the second discourse on philosophical grounding of DSR, arguments are provided for the use of philosophies traditionally used in IS, such as interpretivism or positivism.
- Gregory [14] claims that “DSR is conducted most frequently within a positivistic epistemological perspective”.
- Venable [20] proposes a framework for understanding design research where the framework focuses on theory building as well as evaluation of the solutions from a positivist or interpretivist angle.

Guideline 2: *Understand the philosophical underpinning of research and the discourse on the nature of DSR.*

3. Pragmatism as Paradigm

- March and Smith [10] were some of the first authors to emphasise pragmatism when they argued that truth is what works in practice.
- Hevner in 2007 [22] devoted the closure of his article to claiming pragmatism as the nature of DSR. His view of pragmatism is a 'school of thought that considers practical consequences or real effects to be vital components of both meaning and truth' [22:93].
- A meaningful source on the nature of DSR is the paper by Goldkuhl [25]. In this seminal work he investigates the epistemological foundation for design research and argues that the pragmatist perspective is fit for DSR based on the focus on utility and knowledge growth through development, starting with a problematic situation and aiming for knowledge building.

Guideline 2: ***Understand the philosophical underpinning of research and the discourse on the nature of DSR.***

3. Pragmatism as Paradigm

– Deng, [26] in a recent work, argues that pragmatism is the underpinning philosophy for DSR, but goes through different phases where the researcher is involved as interpretivist, positivist and constructive observer or intervener (Figure 1).

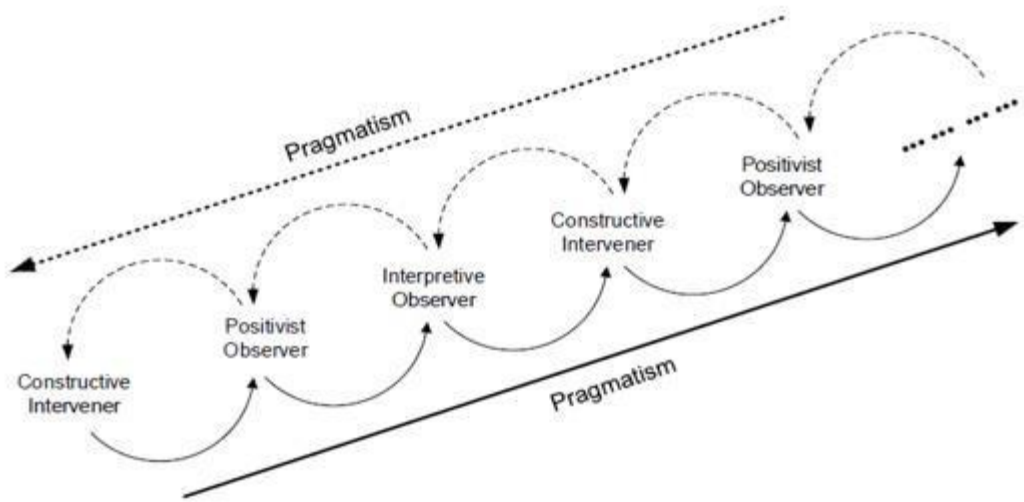

Fig. 1. Iterative design science process [26]

Guideline 3: *Obtain a historical perspective of DSR and consult the work of the pioneers in the field*

- The most frequently cited work in IS is the work of Simon [13, 16] where he argues for the acceptance of the study and development of artificial or man-made objects. He also refers to problems experienced in management and indicates how well-designed systems contributed as artificial objects to solutions in the field of IS.
- Following the work of Simon [13, 16] in terms of the highest number of citations was the seminal work of Hevner et al. [5] in 2004, in which they contrasted behavioural science and DS. In their article, with more than 11 400 citations to date, they presented a framework (Figure 2) for IS research and also provided a set of guidelines for DSR.

Guideline 3: *Obtain a historical perspective of DSR and consult the work of the pioneers in the field*

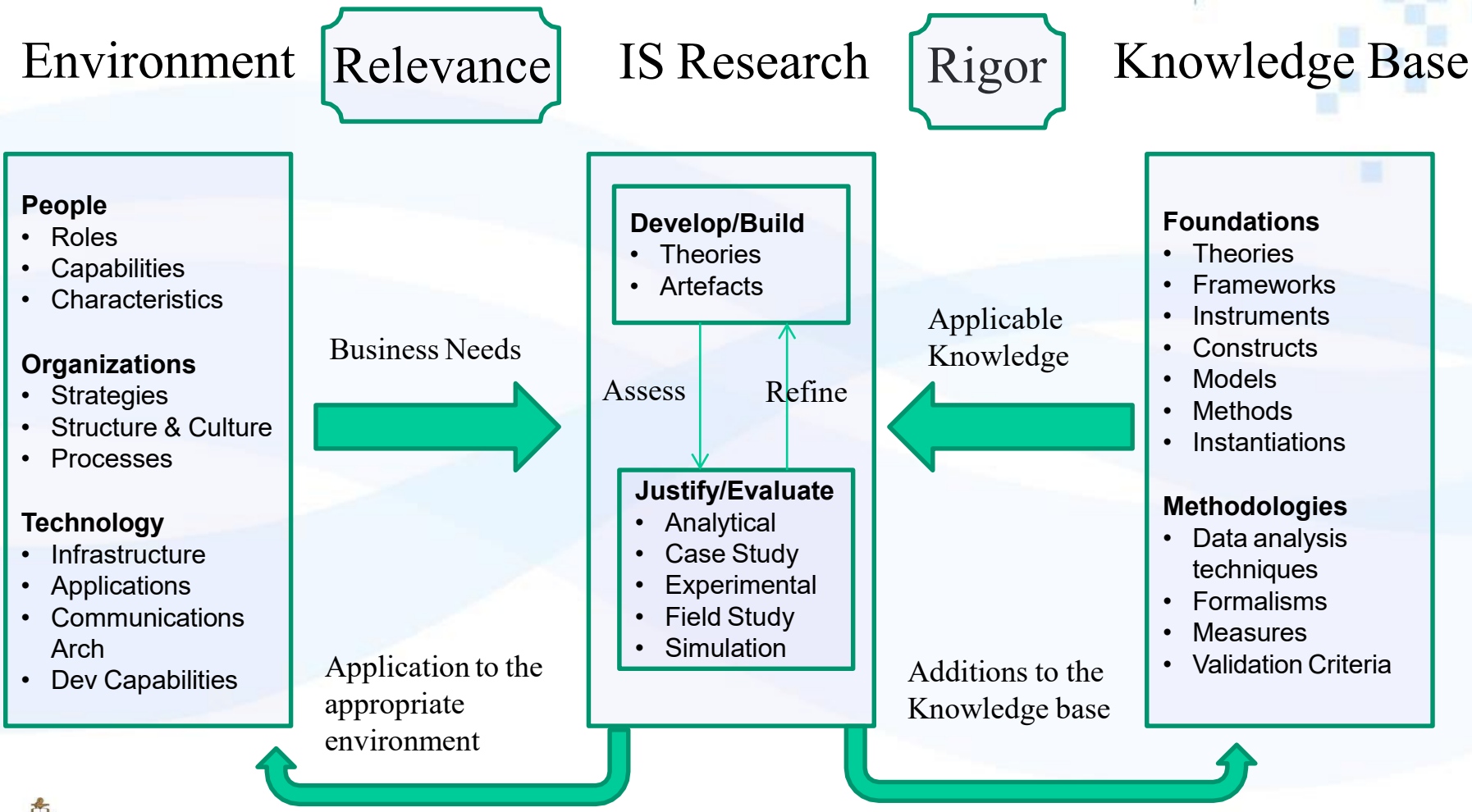

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Table 3. DSR guidelines [5]

Guidelines	
Guideline 1: Design as an Artefact	Design science research must produce a viable artefact in the form of a construct, a model, a method, or an instantiation.
Guideline 2: Problem Relevance	The objective of design science research is to develop technology-based solutions to important and relevant business problems.
Guideline 3: Design Evaluation	The utility, quality, and efficacy of a design artefact must be rigorously demonstrated via well-executed evaluation methods.
Guideline 4: Research Contributions	Effective design science research must provide clear and verifiable contributions in the areas of the design artefact, design foundations, and/or design methodologies.
Guideline 5: Research Rigour	Design science research relies upon the application of rigorous methods in both the construction and evaluation of the design artefact.
Guideline 6: Design as a Search Process	The search for an effective artefact requires utilising available means to reach desired ends while satisfying laws in the problem environment.
Guideline 7: Communication of Research	Design science research must be presented effectively to both technology-oriented and management-oriented audiences.

Guideline 3: *Obtain a historical perspective of DSR and consult the work of the pioneers in the field*

Table 4. Seminal works in DSR [18]

Reference	Cita-tions	Year	Significance
March et al. [10]	3 979	1995	Initially proposed types of artefacts
Walls [2]	1 530	1992	Focuses on design theory - Provides a method for theory building;
Nunamaker et al. [27]	1 508	1991	Proposes a method – argues from the system development background for design
Gregor and Jones [28]	1 428	2007	Focuses on design theory; distinguishes between two different kinds of purposeful artefacts that can be designed: product artefacts and process artefacts.
Gregor and Hevner [29]	1 402	2013	Overview article; positions DSR; gives guidance on publishing DSR;
Cross [19]	1 306	2001	Nature of DSR, distinguishes between scientific design, design science, a science of design
Markus et al. [30]	1 282	2002	Example of a design theory for knowledge management processes

Guideline 4: *Consider the role of the artefact in DSR and the different views on design theory.*

- The first mention of different types of artefacts is by March and Smith [10] as constructs, models, methods and implementations [10].
- In 2003, Rossi and Sein [38] (in acknowledged collaboration with Purao) added 'better theories' as an artefact – however, not all authors agreed.
- Winter [35] gives examples of constructs that include modelling primitives implemented by meta-models of modelling tools, process models implemented as workflows as models and project methods used during software package introduction as a method.
- Purao [36] claimed in 2002 that the artefact created in DSR is software or a system.
- Hevner and Chatterjee [24] and Vaishnavi et al. [17] also give as examples of the artefact algorithms, human/computer interfaces, languages, and system design methodologies.

Guideline 4: *Consider the role of the artefact in DSR and the different views on design theory.*

- In 2010 Offerman et al. [37] did a literature review on the types of artefacts in IS design science and suggested a topology with eight types of artefacts. These included a system design, method, language/notation, algorithm, guideline, requirements, pattern and metric.
- Winter [35], in an editorial, argued that 'although theory building is not design science research, theories as "intermediate" artefacts need to be included in the system of relevant artefacts for IS design science research' [35:472].
- Baskerville [11] emphasised that DSR brings about both practical relevance by developing useful artefacts and scientific rigour by the formulation of design theories.

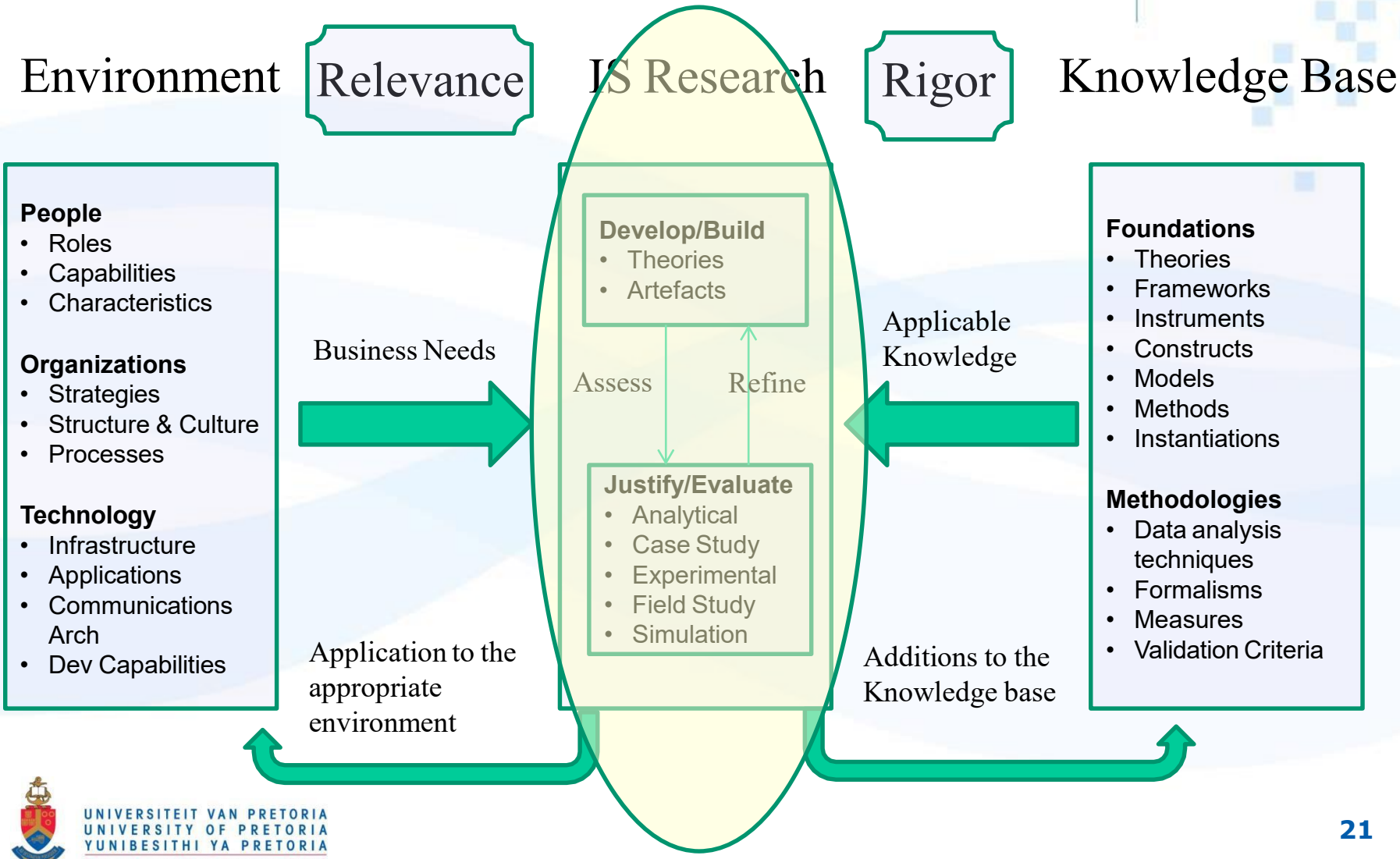
Types of Artefacts in IS

Artefact	Description
Constructs	Constructs can be used in models for example Information Systems Use
Models	A model is a pictorial or graphic representation of key concepts. A model shows, (with the help of arrows and other diagrams), the relationship between various types of variables e.g. independent, dependent, moderating, mediating variables etc.
Methods	A particular procedure for accomplishing or approaching something, especially a systematic or established one.
Implementations	The process of putting a decision or plan into effect; execution.
Design theory	An **Design Theory** is a prescriptive **theory** that offers **theory**-based principles, which can guide practitioners and scholars in the **design** of effective **systems** and set an agenda for on-going research.

Guideline 5: *Select an appropriate DSR method for execution of the research study*

- March and Smith [10] argued that design science consists of two basic activities, namely building and evaluating.

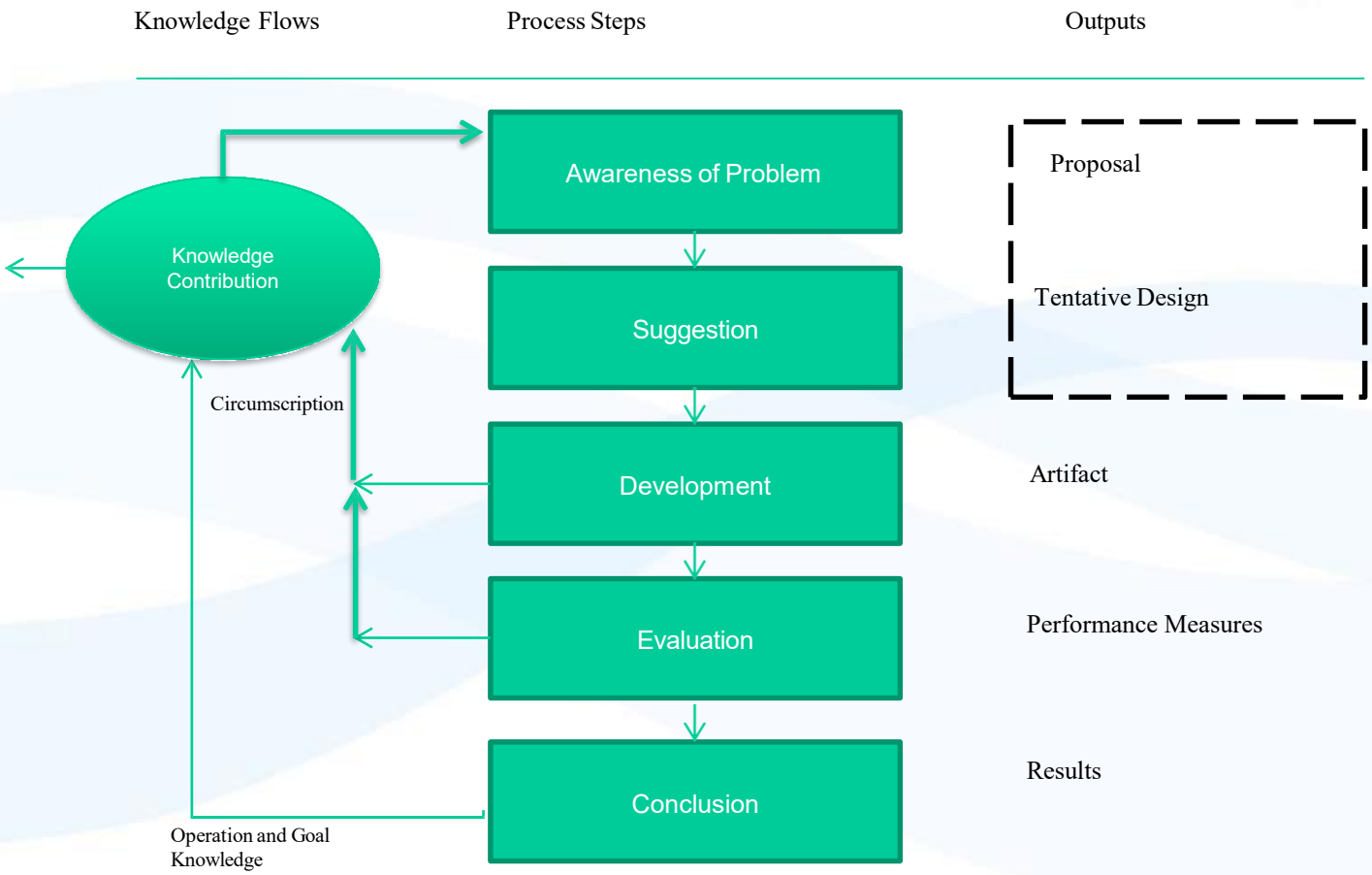
Hevner et al [1]



Guideline 5: *Select an appropriate DSR method for execution of the research study*

- March and Smith [10] argued that design science consists of two basic activities, namely building and evaluating.
- All the methods shared in the literature on conducting DSR consist of a combination of the general design and development phases, namely identification, design, development and testing.
- Vaishnavi et al. [17] published one of the often used and referenced methods that they call a DSR process model, which was based on work from Takeda et al. [12].

Vaishnavi [2]



http://desrist.org/desrist/content/design-science-research-in-information-systems.pdf

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- Another popular DSR process model used by authors is the work published by Peffers et al. [42]. In their process model (Figure 4) the DSR cycles through problem identification and motivation, objectives of solution, design and development, demonstration, evaluation and communication.

Peffers [15]

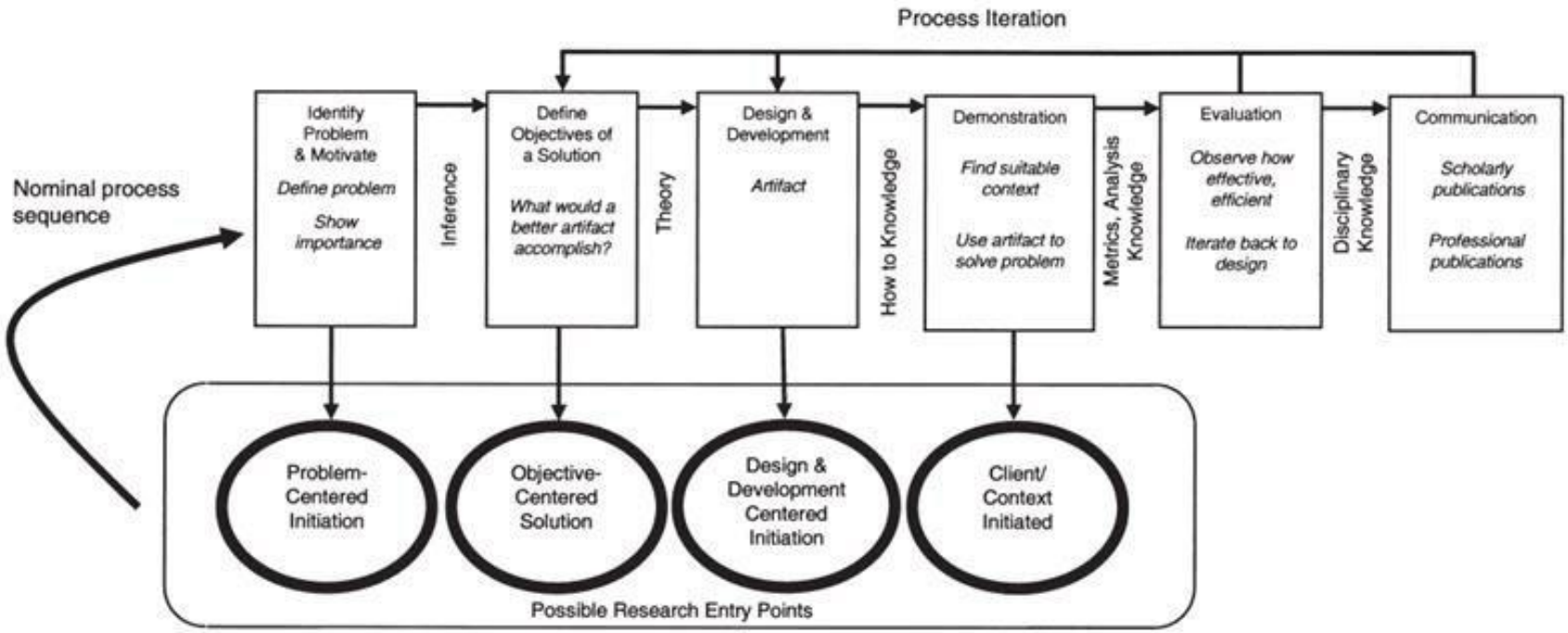


Figure 1. DSRM Process Model

Guideline 5: *Select an appropriate DSR method for execution of the research study*

- For the evaluation of the artefact, the pioneers working in this field were Pries-Heje, Baskerville and Venable, who published a number of articles [20, 48, 49] building up to a framework for evaluation in design science (FEDS) [50].
- In their framework they identified four different possible strategies, namely the quick and simple strategy, the human risk and effectiveness evaluation strategy, the technical risk and efficacy evaluation strategy, and the purely technical artefact strategy.
- They then continued to provide a four-step process for choosing an approach for a particular DSR, namely 1) explicate the goals of the evaluation, 2) choose the evaluation strategy or strategies, 3) determine the properties to be evaluated, and 4) design the individual evaluation episode(s).

Guideline 6: *Strategise on how research done in DSR should be communicated in a report such as a thesis.*

- Gregor and Hevner [29] give an overview on publishing an article on DSR in which they propose a publication schema for recording results.
- They argue that the four questions that reviewers will ask are whether:
  - the problems discussed in the paper are of substantial interest,
  - whether the problems are solved or a contribution is made to a solution,
  - whether the methods are new and
  - whether the paper supports understanding of the area of research.

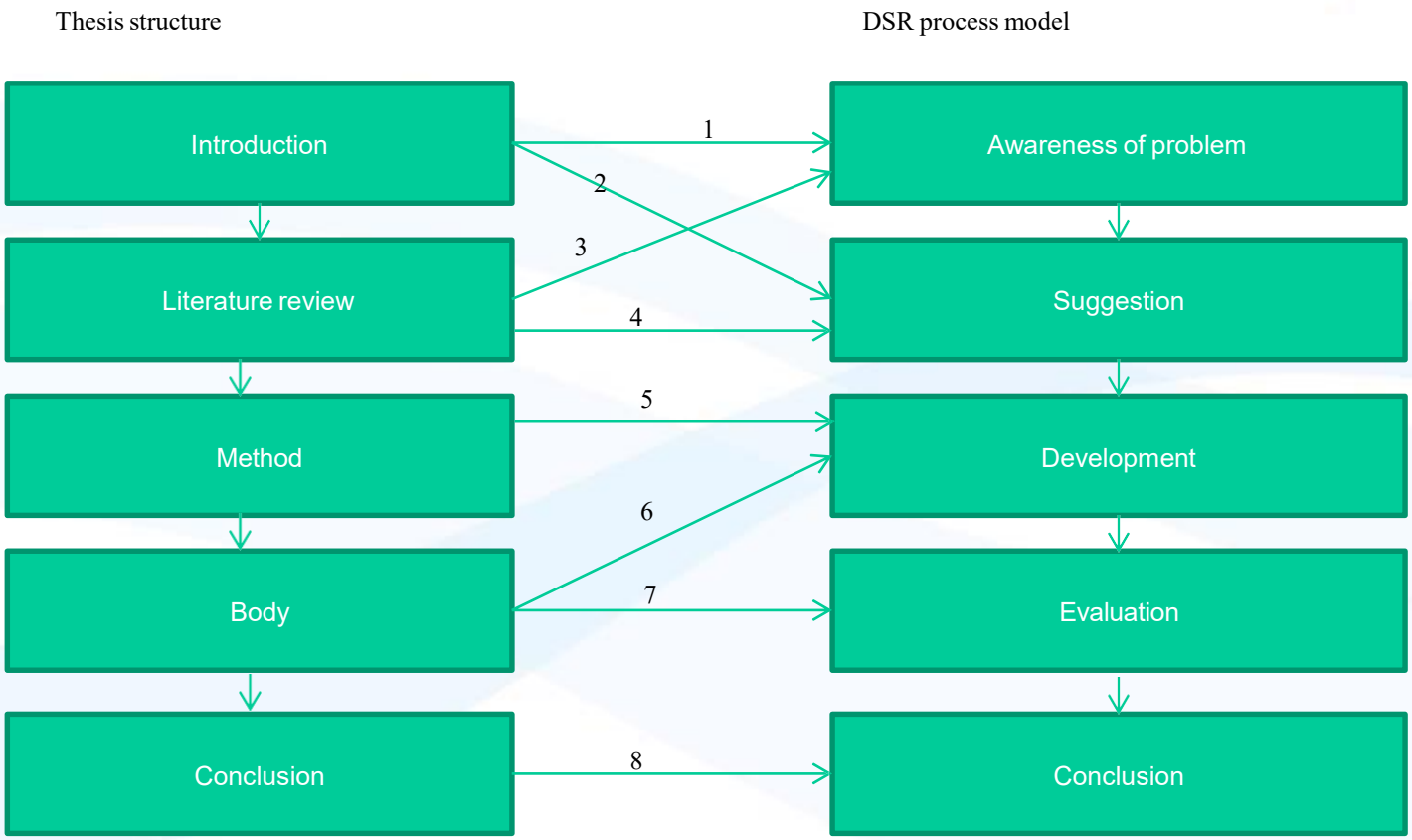
Guideline 6: *Strategise on how research done in DSR should be communicated in a report such as a thesis.*

- Van der Merwe et al. [55] describe a method that a student can use to write a thesis according to the steps proposed by Vaishnavi et al. [17].
- They argue that:
  - the introduction and literature review of a thesis maps to the awareness phase,
  - the literature review and research design map to the suggestion phase,
  - the research design and body of the thesis give an overview of the development,
  - while the body of the thesis should also report on the evaluation phase.
  - The last phase, the conclusion, will then also be handled in the conclusion of the thesis.

Guideline 5: *Strategise on how research done in DSR should be communicated in a report such as a thesis.*

Van der Merwe, A., Gerber, A., Smuts, H.: Mapping a Design Science Research Cycle to the Postgraduate Research Report. In: Liebenberg, J. and Gruner, S. (eds.) ICT Education. pp. 293–308. Springer International Publishing, Cham (2017). https://doi.org/10.1007/978-3-319-69670-6_21.

Structure of a research report: Scenario 1 – Single DSR cycle



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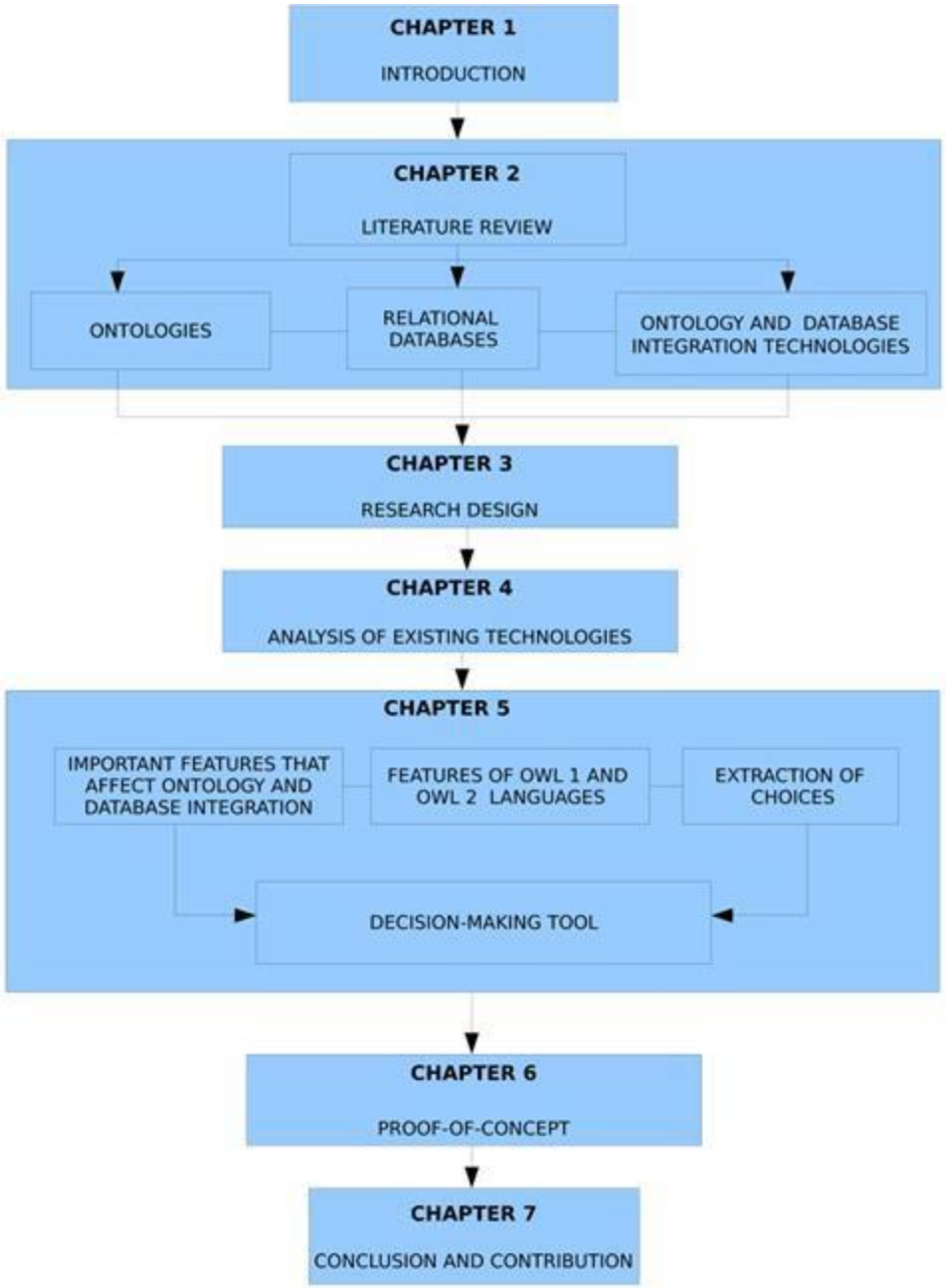
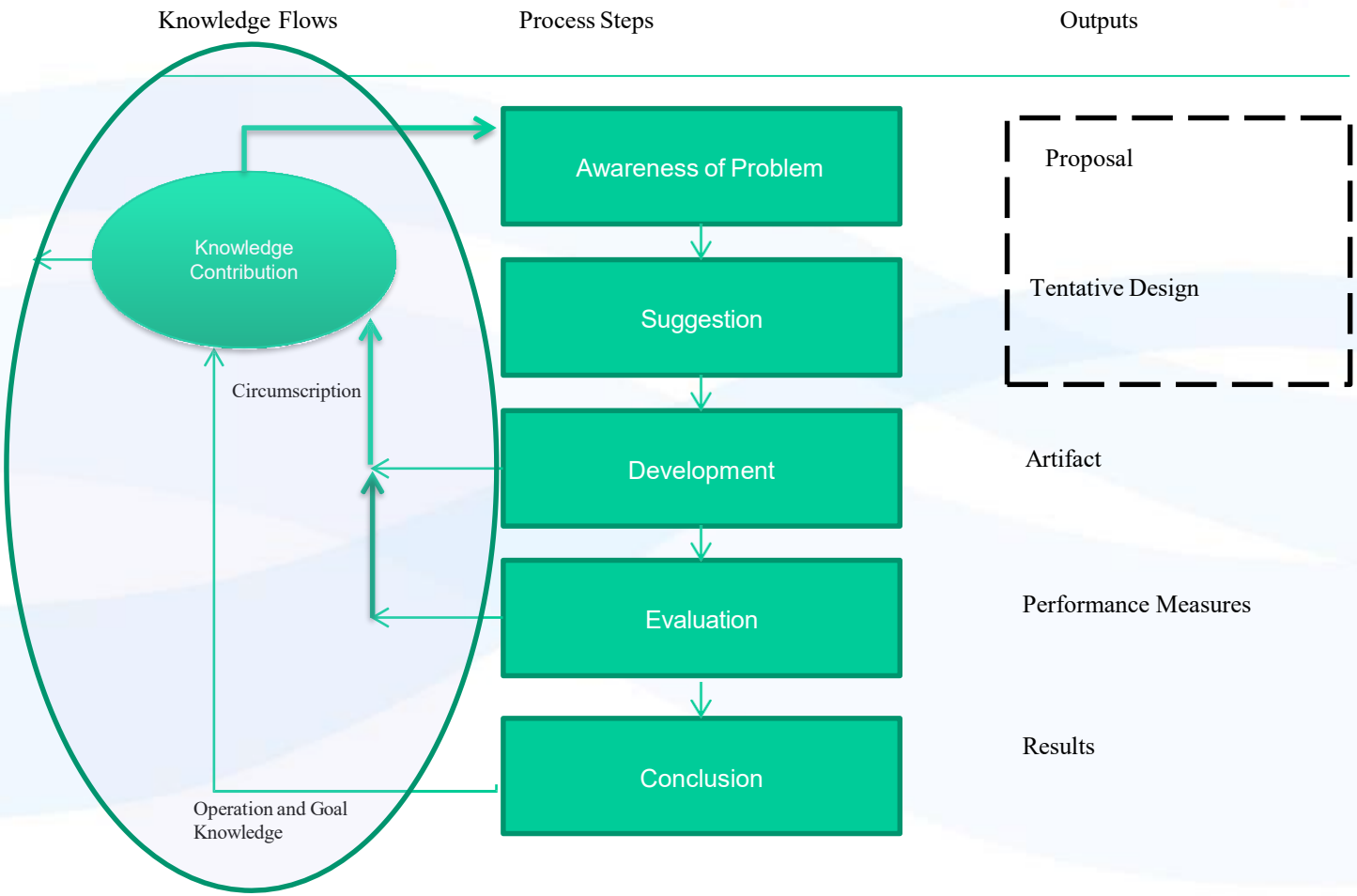
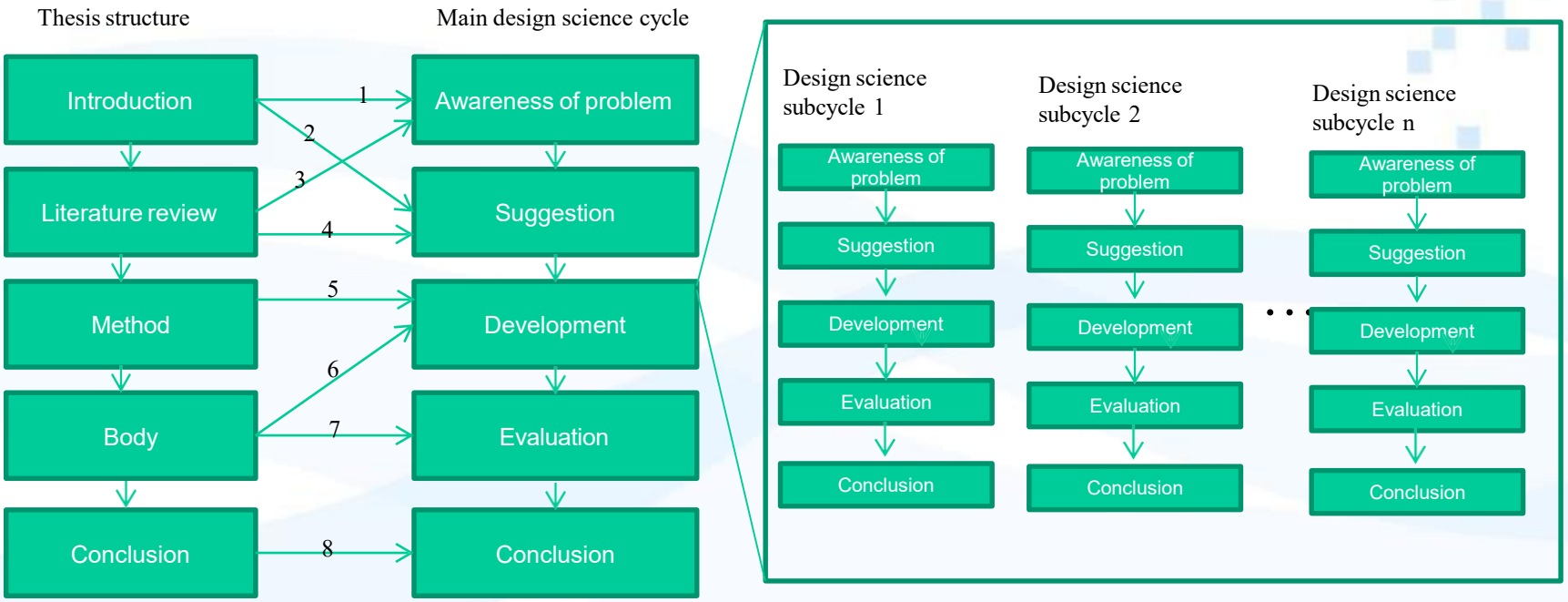

CHAPTER 1
INTRODUCTION
CHAPTER 2
LITERATURE REVIEW
ONTOLOGIES
RELATIONAL DATABASES
ONTOLOGY AND DATABASE INTEGRATION TECHNOLOGIES
CHAPTER 3
RESEARCH DESIGN
CHAPTER 4
ANALYSIS OF EXISTING TECHNOLOGIES
CHAPTER 5
IMPORTANT FEATURES THAT AFFECT ONTOLOGY AND DATABASE INTEGRATION
FEATURES OF OWL 1 AND OWL 2 LANGUAGES
EXTRACTION OF CHOICES
DECISION-MAKING TOOL
CHAPTER 6
PROOF-OF-CONCEPT
CHAPTER 7
CONCLUSION AND CONTRIBUTION

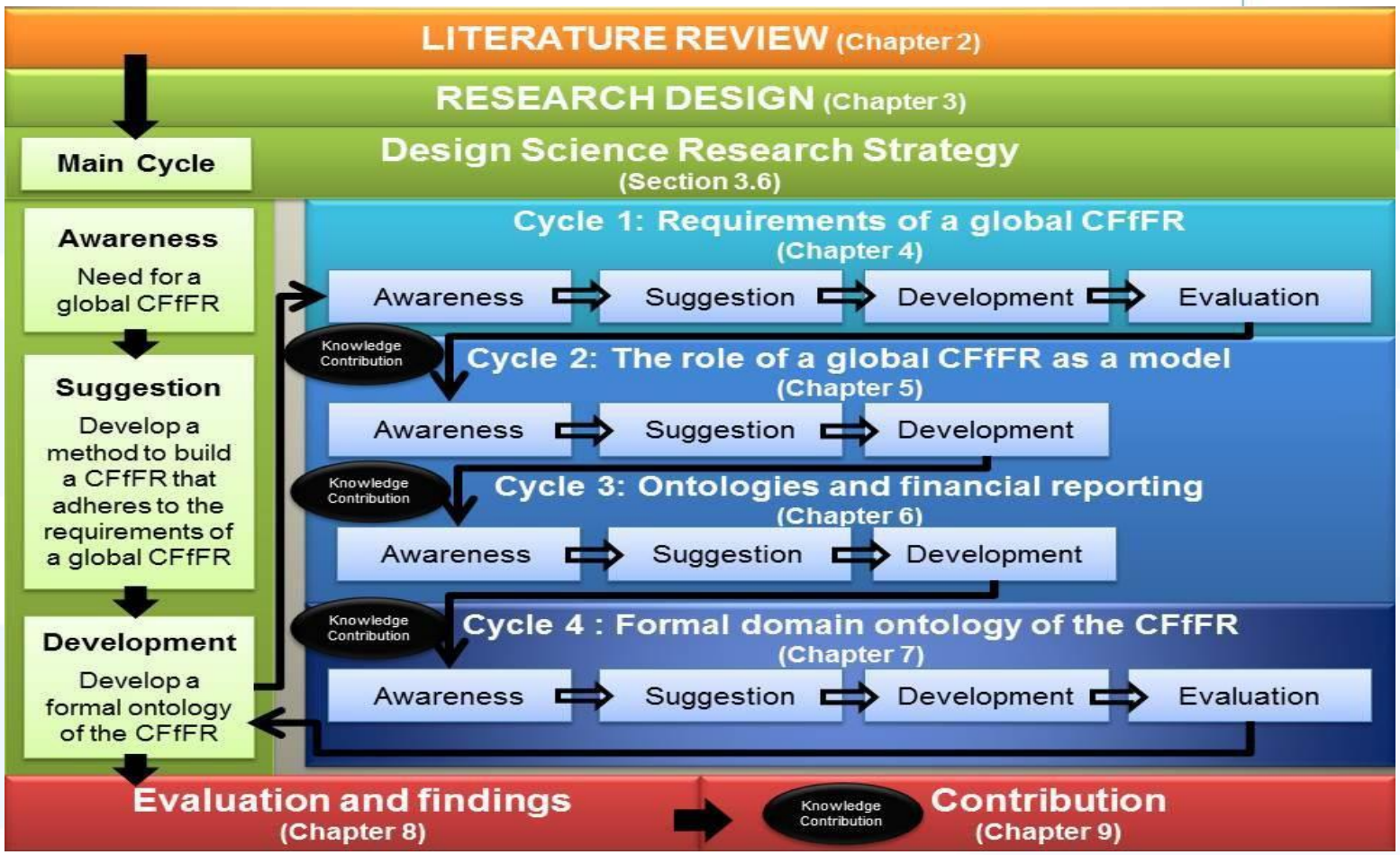
Vaishnavi [2]

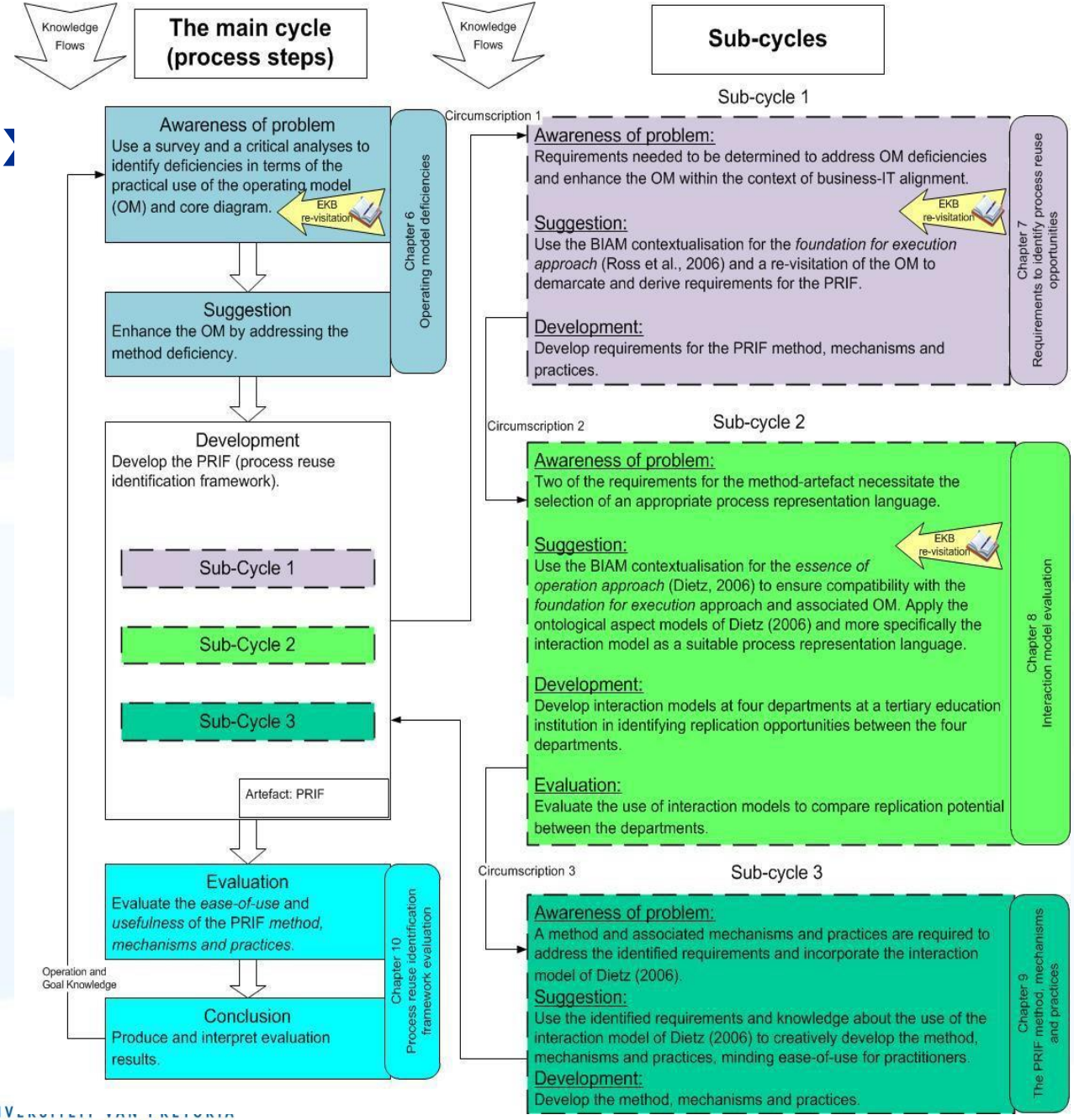

Knowledge Flows
Process Steps
Outputs
Knowledge Contribution
Circumscription
Operation and Goal Knowledge
Awareness of Problem
Suggestion
Development
Evaluation
Conclusion
Proposal
Tentative Design
Artifact
Performance Measures
Results

Structure of a research report: Scenario 2: Multiple cycles of design



Example: Scenario 2




Knowledge Flows
The main cycle (process steps)
Knowledge Flows
Sub-cycles
Awareness of problem
Use a survey and a critical analyses to identify deficiencies in terms of the practical use of the operating model (OM) and core diagram.
EKB re-visitation
Chapter 6 Operating model deficiencies
Suggestion
Enhance the OM by addressing the method deficiency.
Development
Develop the PRIF (process reuse identification framework).
Sub-Cycle 1
Sub-Cycle 2
Sub-Cycle 3
Artefact: PRIF
Evaluation
Evaluate the ease-of-use and usefulness of the PRIF method, mechanisms and practices.
Chapter 10 Process reuse identification framework evaluation
Operation and Goal Knowledge
Conclusion
Produce and interpret evaluation results.
Circumscription 1
Sub-cycle 1
Awareness of problem:
Requirements needed to be determined to address OM deficiencies and enhance the OM within the context of business-IT alignment.
EKB re-visitation
Suggestion:
Use the BIAM contextualisation for the foundation for execution approach (Ross et al., 2006) and a re-visitation of the OM to demarcate and derive requirements for the PRIF.
Development:
Develop requirements for the PRIF method, mechanisms and practices.
Chapter 7 Requirements to identify process reuse opportunities
Circumscription 2
Sub-cycle 2
Awareness of problem:
Two of the requirements for the method-artefact necessitate the selection of an appropriate process representation language.
EKB re-visitation
Suggestion:
Use the BIAM contextualisation for the essence of operation approach (Dietz, 2006) to ensure compatibility with the foundation for execution approach and associated OM. Apply the ontological aspect models of Dietz (2006) and more specifically the interaction model as a suitable process representation language.
Development:
Develop interaction models at four departments at a tertiary education institution in identifying replication opportunities between the four departments.
Evaluation:
Evaluate the use of interaction models to compare replication potential between the departments.
Chapter 8 Interaction model evaluation
Circumscription 3
Sub-cycle 3
Awareness of problem:
A method and associated mechanisms and practices are required to address the identified requirements and incorporate the interaction model of Dietz (2006).
Suggestion:
Use the identified requirements and knowledge about the use of the interaction model of Dietz (2006) to creatively develop the method, mechanisms and practices, minding ease-of-use for practitioners.
Development:
Develop the method, mechanisms and practices.
Chapter 9 The PRIF method, mechanisms and practices

Questions?